

Run basic STA on synthesized netlist using timing libraries or representative delays.

Project Report

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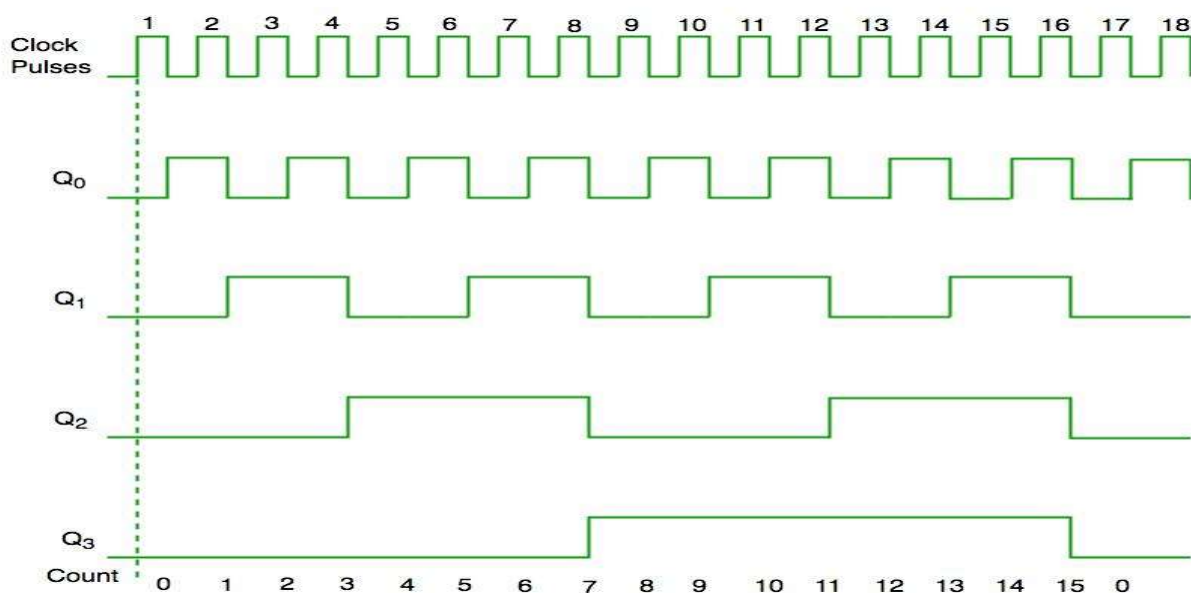
Abstract

This project demonstrates the complete flow of synthesizing a 4-bit synchronous counter, applying timing constraints, and performing Static Timing Analysis (STA). The objective is to verify that the synthesized gate-level netlist meets setup and hold timing requirements under a specified clock frequency. STA was performed using representative timing delays and timing constraints. Timing reports were analyzed to identify any violations and suggest possible fixes.

Introduction

Static Timing Analysis (STA) is a method used in digital design to verify timing correctness without applying simulation vectors. STA checks whether signals can propagate through the design within the required clock period while satisfying setup and hold constraints.

In this project, a 4-bit synchronous counter is used as the example design. The counter increments its value on every rising edge of the clock and is reset asynchronously.



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Design Specification

Parameter	Value
Design	4-bit Synchronous Counter
Clock Frequency	100 MHz
Clock Period	10 ns
Reset Type	Asynchronous Reset
Implementation	Verilog RTL

RTL Design

The counter value `q` increments by one on every rising edge of `clk`.
When `rst` is asserted, the counter resets to zero.

Synthesis Flow

The RTL was synthesized into a gate-level netlist using a synthesis tool such as Xilinx Vivado or an equivalent open-source flow.

Synthesis Steps

1. Create a new project.
2. Add the Verilog source file.
3. Add the timing constraints file.
4. Run synthesis.

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5. Generate the synthesized netlist.

Timing Constraints (SDC/XDC)

These constraints define the clock period and external input/output timing assumptions used by STA.

Static Timing Analysis Procedure

After synthesis, STA was executed using the synthesized netlist and timing constraints.

Example Vivado TCL Commands

Example OpenSTA Commands

Timing Analysis Results

Example Setup Timing Report

Interpretation: The maximum combinational delay between registers is 5.782 ns, which is less than the required 10 ns clock period. Therefore, setup timing is satisfied.

Example Hold Timing Report

Interpretation: Positive hold slack indicates that data remains stable long enough after the clock edge. Hold timing is satisfied.

Worst Negative Slack (WNS)

Since WNS is positive, the design has no setup timing violations.

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Critical Timing Path

The longest path in the counter is typically:

This path determines the maximum operating frequency of the design.

Timing Violation Scenario

If the clock period is reduced aggressively, timing violations can occur.

Example

Suppose the resulting report shows:

This negative slack indicates that the data path cannot complete within 2 ns, causing a setup violation

Methods to Fix Timing Violations

1. Reduce Clock Frequency

Increase the clock period, for example from 2 ns to 10 ns.

2. Pipelining

Insert additional registers to reduce combinational delay per stage.

3. Logic Optimization

Simplify arithmetic logic and reduce fanout.

4. Re-synthesis with Optimization

Use synthesis optimization directives such as:

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Final Results Summary

Parameter	Result
Clock Period	10 ns
Data Path Delay	5.782 ns
Setup Slack	4.218 ns
Hold Slack	0.612 ns
Worst Negative Slack (WNS)	4.218 ns
Timing Status	PASSED

Conclusion

Static Timing Analysis was successfully performed on the synthesized 4-bit synchronous counter netlist. Timing constraints were applied, and both setup and hold checks were verified. The design achieved positive setup and hold slack values, indicating correct timing operation at 100 MHz. A hypothetical timing violation scenario was also analyzed, and corrective techniques such as pipelining, logic optimization, and clock-period adjustment were discussed.

This project demonstrates the basic STA flow used in digital IC and FPGA design to validate timing closure before implementation or fabrication.

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References

1. Xilinx Vivado User Guide.
2. Static Timing Analysis fundamentals in digital design.
3. OpenSTA documentation and Liberty timing model format.